Q–113

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BCA II Sem Main/Back Examination 2025-26

BCA–204 - Data Structure

Time : 3 Hours] [Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to 25 words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part–B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of the following supporting materials is permitted during examination. (Mentioned in form no. 205).

1 NIL **2** NIL

Part–A **(10 × 2 = 20)**

1. What is an Abstract Data type ?
2. What is sparse matrix ? Give a real life example.
3. How are strings represented in C ?
4. Why is linear search preferred over binary search ?
5. What is stack overflow condition ?
6. What is the time complexity of Push() and Pop() in a stock ?
7. What is the use of '*' and '&' operators in pointer operations ?
8. What is doubly linked list ?
9. How can a tree be represented in memory ?
10. Define Vertex and edge.

Part–B **[4 × 5 = 20]**

1. Compare the time and space complexity of polynomial addition using array representations.
2. Compare recursive and iterative binary search in terms of memory usage and performance.
3. What is stack and how is it implemented using dynamic arrays ?
4. How is circular queue implemented using dynamic arrays ?
5. Why is dynamic memory use for implementing stacks and queues ?
6. What is the difference between malloc() and calloc() with suitable example ?
7. What are the advantages of linked representation of binary trees ?

Part–C **[3 × 10 = 30]**

1. Design a self-referential structure in C to represent a polynomial. How does it improve over a simple array-based representation ?

2. Calculate the time complexity of the naive pattern matching algorithm with example.

3. What are the steps to convert an infix expression to postfix expression using a stack ?

4. Define a singly linked list. How is a singly linked list created and traversed ? Write a C program to insert a node at the beginning middle and end of a singly linked list and also explain how deletion in a singly linked list and explain how BFS can be used to find the shortest path in an unweighted graph.

5. Describe the Breadth-First Search (BFS) algorithm. How is it implemented using a queue. Provide an example and explain how BFS can be used to find the shortest path in an unweighted graph.

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